

	A	B	C	D
1	230	200	210	240
2	190	210	200	200
3	200	180	240	220
4	220	180	210	230

a)

30 0 10 40
0 20 10 10
20 0 60 40
40 0 30 50

~~30 0 0 30~~
~~0 20 0 0~~
20 0 50 30
40 0 20 40

30 20 0 30
0 40 0 0
0 0 30 10
20 0 0 20

1-C 210
2-D 200
3-A 200
4-B 180
Z = 790

b)

	A	B	C	D
Fict	0	0	0	0
2	190	210	200	200
3	200	180	240	220
4	220	180	210	230

~~0 0 0 0~~
0 20 10 10
20 0 60 40
40 0 30 50

~~10 10 0 0~~
~~0 20 0 0~~
20 0 50 30
40 0 20 40

10 30 0 0
0 40 0 0
0 0 30 10
20 0 0 20

2-D 200 2-A 190
3-A 200 3-B 180
4-B 180 4-C 210
C fica sem a peça D fica sem
z = 580 580

c)

2-C 200
3-A 200
4-B 180
D fica sem a peça
z = 580

d)

	A	B	C	D
1	230	200	210	240
2	190	210	200	200
3	200	180	240	220
4	220	180	210	230

10	0	0
0	60	40
0	30	50

40	0	0
0	30	10
0	0	20

1-A 230
2-D 200
3-B 180
4-C 210
z= 820

e)

	A	B	C	D
1	230	200	210	240
2	190	210	200	200
3	M	180	240	220
4	220	180	210	230

30	0	10	40
0	20	10	10
M-180	0	60	40
40	0	30	50

30	0	0	30
0	20	0	0
M-180	0	50	30
40	0	20	40

30	20	0	30
0	40	0	0
M-200	0	30	10
20	0	0	20

20	20	0	30
0	50	10	0
M-210	0	30	0
10	0	0	10

1-C 210
2-A 190
3-D 220
4-B 180
z= 800